Title: The impact of sound feedback on long-term posture correction to evaluate if simple audio alerts can successfully build healthier habits.

Background

Prolonged computer use induces "habit decay," where cognitive task focus causes users to lose conscious awareness of slouch. Real-time auditory biofeedback is a potential solution, but its ability to train lasting muscle memory after the feedback is deactivated remains unquantified. This study investigated if real-time webcam-based audio biofeedback can successfully build muscle memory to maintain healthy sitting alignment.

Methods

Using a randomized controlled design, 8 participants completed a 20-minute standardized typing task followed by a 5-minute feedback-free retention phase. A webcam-based application tracked nose and shoulder coordinates at 10Hz using Google MediaPipe. Sinking 15° below a calibrated upright baseline triggered a slouch violation. During the first 20 minutes, the Active group (N=4) received a 1000Hz tone reminder upon violation, while the Control group (N=4) was monitored silently. During the final 5 minutes, the tone was deactivated for both groups to evaluate postural muscle memory retention.

Results

During the initial 20-minute phase, the Control group spent an average of 36.0 seconds in the slouch zone, while the Active group, utilizing biofeedback, spent only 18.5 seconds slouching, a 48.6% reduction. In the subsequent 5-minute retention phase (with all feedback disabled), the Control group spent an average of 9.0 seconds slouching. In contrast, the Active group maintained significant postural correction, spending an average of only 3.3 seconds in the slouch zone. This represents a 63.3% relative reduction in slouch duration post-intervention.

Conclusion

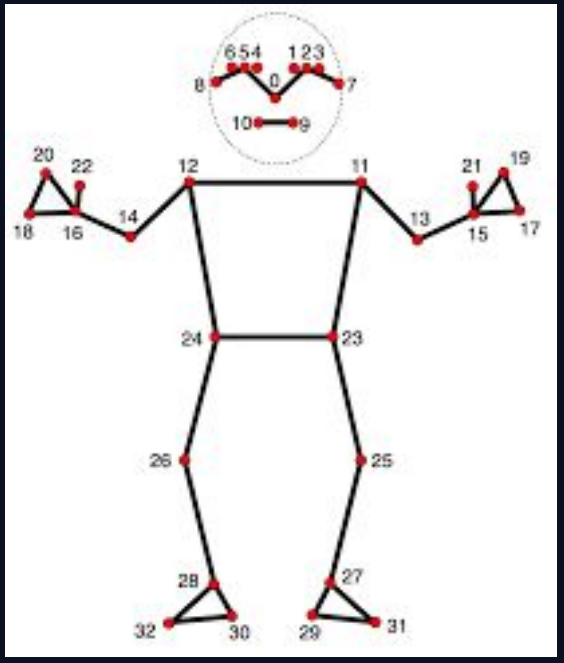
Webcam-based audio biofeedback does not act merely as a temporary reminder; it successfully trains lasting postural muscle memory, significantly reducing slouch duration even after the auditory stimulus is removed.

Kinetic Biofeedback & Postural Decay

The impact of sound feedback on long-term posture correction to evaluate if simple audio alerts can successfully build healthier posture habits.

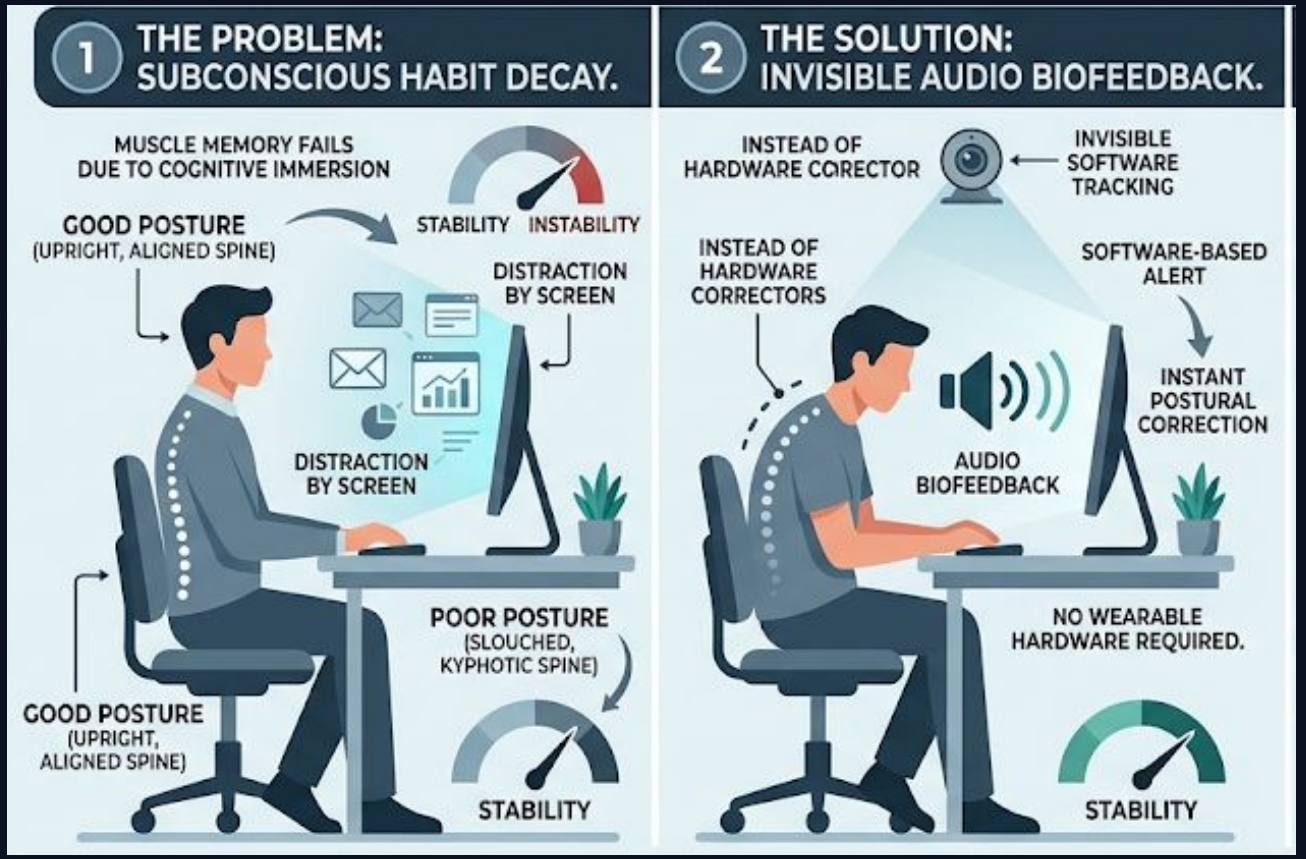
Presenter: **Umang Patel**

PI: **Dr. Bhounsule**



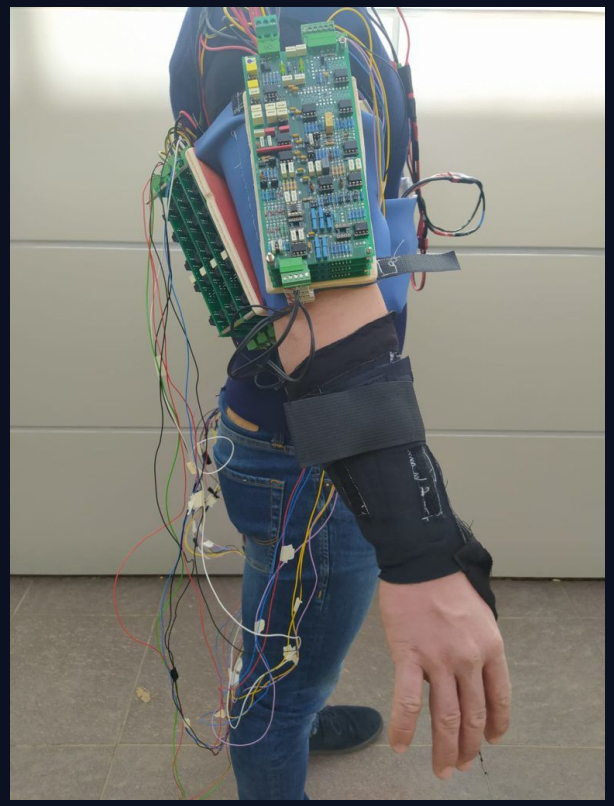
The Problem: Tech Neck

- Deep focus causes posture neglect.
- Leads to severe neck strain.
- Causes permanent spinal misalignment.
- Unnoticed until chronic pain begins.
- Current wearables are too intrusive.

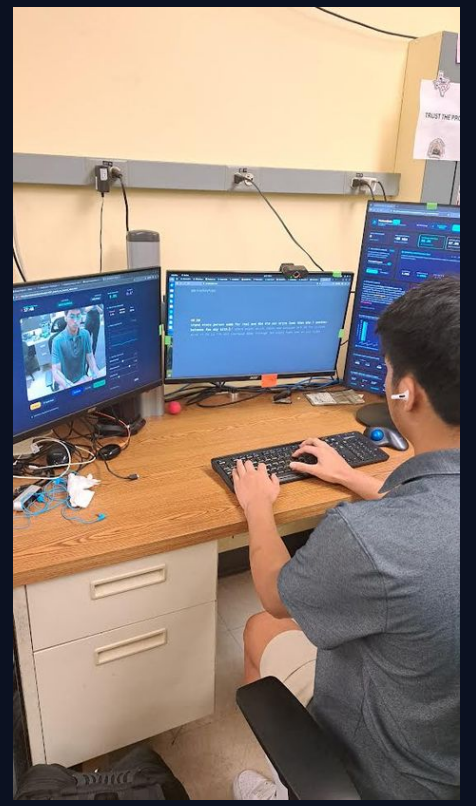


Background & Current Limits

- Wearable IMU sensors are expensive.
- Physical markers are highly uncomfortable.
- My solution uses purely software.
- Conscious postural loop-the cycle of an individual being aware of their body's physical alignment



V/S



Hypothesis

"Audio-visual biofeedback triggered by slouching thresholds will significantly reduce the total time spent slouching and disrupt habit decay"



Posture Drops

User subconsciously slouches past the 15° threshold.



Audio Alert

Immediate 1000 Hz sound accurately triggers.



Habit Reset

Subconscious posture correction instantly occurs.

Trial Design: Experimental Phases

A 25-minute study mapping the transition from training to retention.

PHASE 1 — **20 MINUTES**

Active Phase

Active Group (n=4)
Receives **Real-Time Audio Feedback** (1000Hz beep) immediately upon slouching >15° from calibration.

Control Group (n=4)
Subjected to **Silent Monitoring**. Posture tracked but no alerts provided.

PHASE 2 — **5 MINUTES**

Retention Phase

Audio Deactivated. Tracking continues silently to measure muscle memory retention.

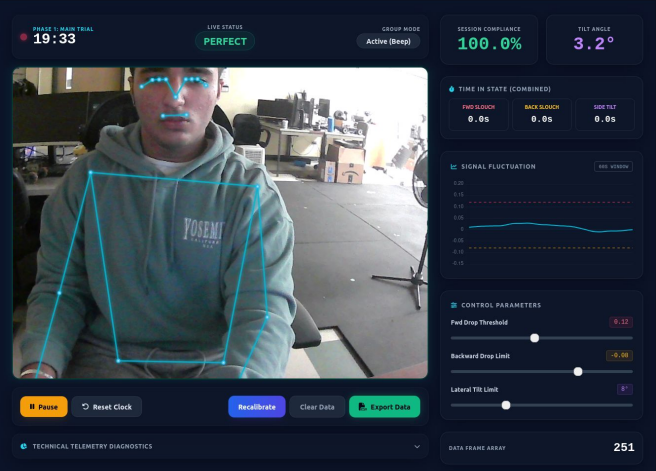
Control Group (n=4)
Monitoring continues **Silently**. Serves as the natural habit decay baseline.

Study Design

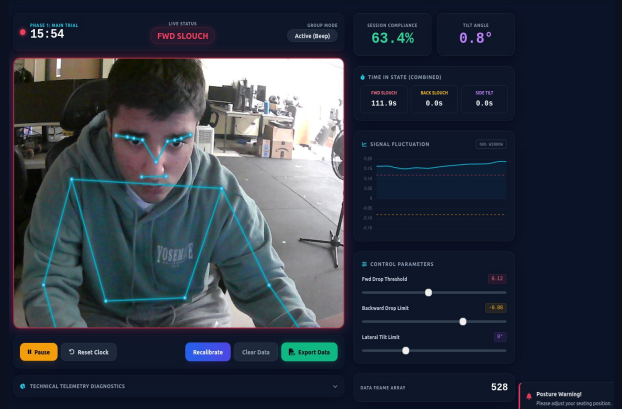
Parameter	Control	Active
Feedback State	Silent Passive Logging	1000 Hz Audio Biofeedback
Trigger Logic	None (Baseline Decay)	Slouch > 15° for > 3.0s

Environmental Protocol

- Standardized 30 inches desk height
- Standardised 27 inches chair height
- Monkey Type
- Primary Metric- Time in "Ideal" Range
- Session Length- 25 minutes
- Sampling Rate: 30 FPS



v/s



Methodology & Tracking

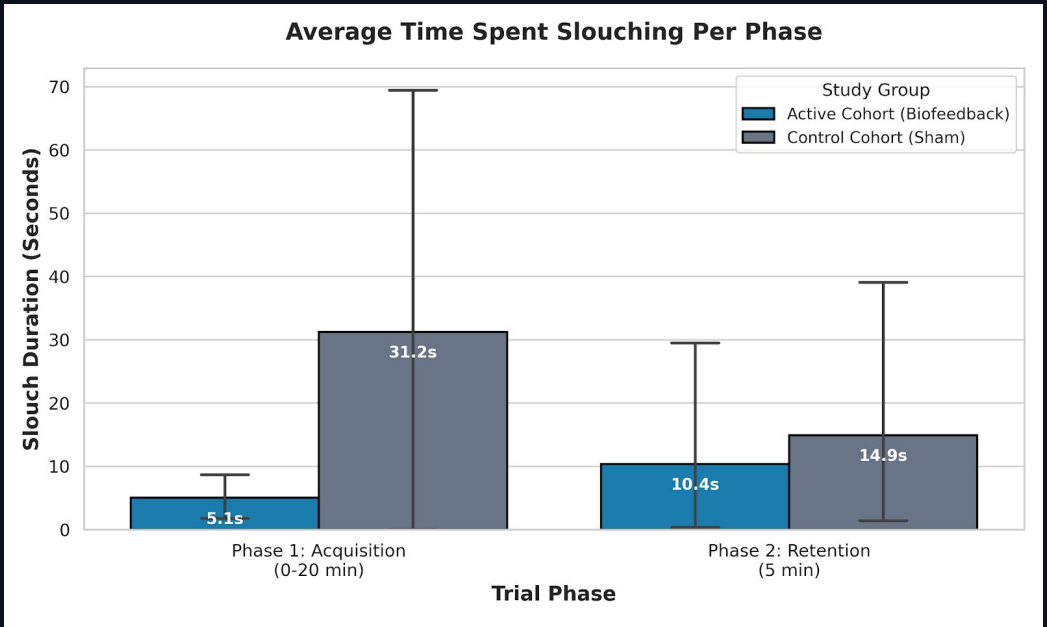
MediaPipe -Landmark Pose

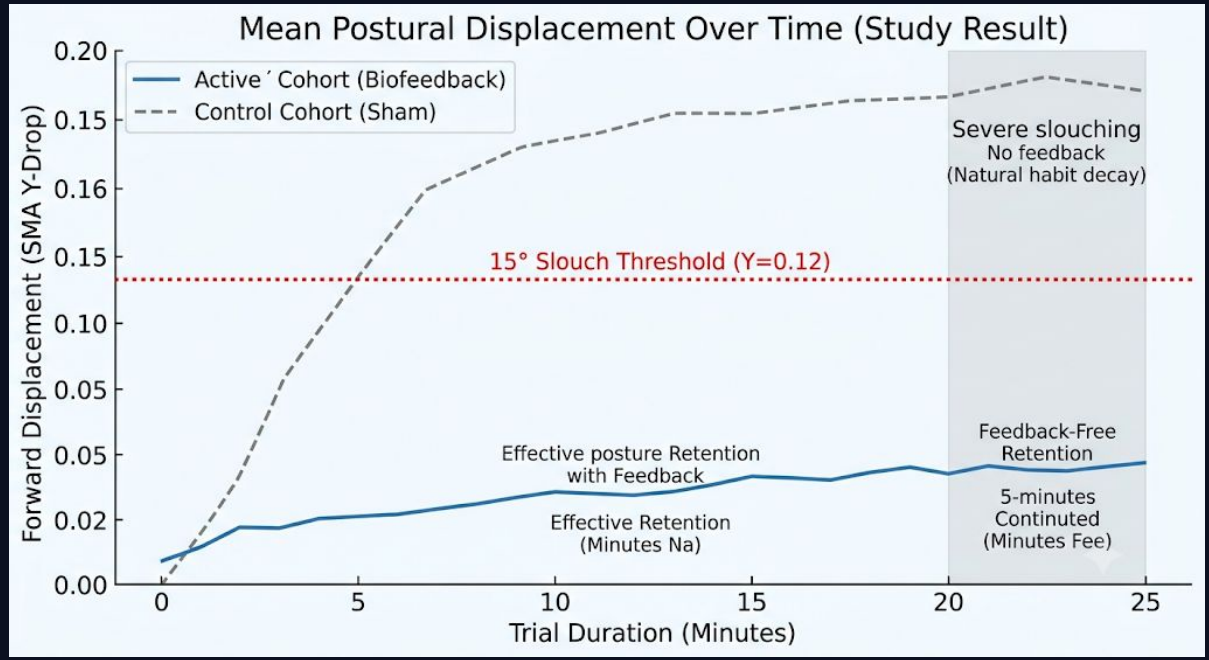
- **Calibration:** Establish "True Upright" baseline coordinates using computer vision upon session start.
- **Collection:** Extract nose and shoulder coordinates at 10 Hz via standard webcam.
- **Depth Inference:** Algorithm synthetically calculates depth (Z-axis) by analyzing pixel size and perspective ratios of key landmarks.
- **Conditioning:** Active group receives immediate 1000 Hz alert upon breaching threshold.

Signal Processing

To eliminate natural jitter from the webcam feed, a digital low-pass filter with a cutoff frequency of 1Hz is applied to the raw kinematic data.

Habit Decay vs. Biofeedback





Phase 1: Active (0–20 min)

- Active : Averaged only 5.0 seconds of slouching.
- Control : Averaged 31.2 seconds of slouching.

Phase 2: Retention (5 min)

- Active : Averaged 10.4 seconds of slouching.
- Control : Averaged 14.9 seconds of slouching.

Insight: The active group maintained better posture than the control group even after the biofeedback was removed, indicating a short-term learning or retention effect.

Active Correction (Active Phase):

- Trend: The active group's slouching remains consistently low and nearly flat throughout the 20-minute training period.

Short-Term Learning (Retention Phase):

- Trend: Once the biofeedback is turned off, the active group's posture begins to slowly degrade, but at a heavily reduced rate compared to the control group.

Core Conclusion: Biofeedback doesn't just act as a temporary fix. It actively trains the user. The data shows clear evidence of short-term learning, where users subconsciously maintain better posture even after the device is silenced.

QUANTITATIVE EFFICACY

Active Correction

Trend: The active group's slouching remains consistently low and nearly flat throughout the 20-minute training period.

Short-Term Learning

Trend: Once biofeedback is off, active group's posture degrades at a heavily reduced rate compared to control group.

48.6%

Training Reduction

Amount of time saved from slouching during active phase.

63.3%

Retention Lead

Performance improvement over the silent control group.

Core Conclusion: Biofeedback doesn't just act as a temporary fix. It actively trains the user.

Conclusions & Future Direction

Primary Findings

- **Interruption of Decay:** Auditory cues successfully interrupt cognitively-induced muscle relaxation by resetting the conscious postural loop.

Future Directions

- **Scale Testing:** Expand sample size to validate algorithm across diverse body types and desk setups.
- **Integration:** Develop a low-overhead background application for continuous workplace monitoring.

Thank You



PROJECT MENTORS & TEAM

- **Nathan Morabito**
- **Dr. Pranav Bhounsule**
- **Dr. Zhinan Wang**
- **Dr. Amelia Bartholomew**
- The Robotics and Motion Learning Lab



OUR VALUED TEST SUBJECTS

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- Grant Bowman
- Hunter Morgan
- Lainey Laughton
- Stephen Stolyarov
- Daphne Gil